THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Electronic and Computer Engineering
ELEC 1100

Laboratory 4: Transistor and Motor Interface (5%)

A) Objectives:

- To study transistor characteristics.
- To control the DC motor with ICs.

B) Equipment:

- Bipolar junction transistor (NPN: PN2222A)
- DC voltage regulator (LM7805), DC motors, Dual H-Bridge Motor Driver IC (L293D), Hex Schmitt−Trigger Inverter (74HC14)

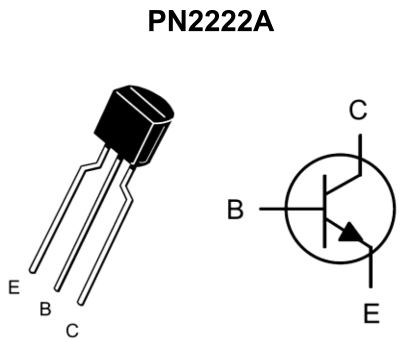
C) Experiment Procedures:

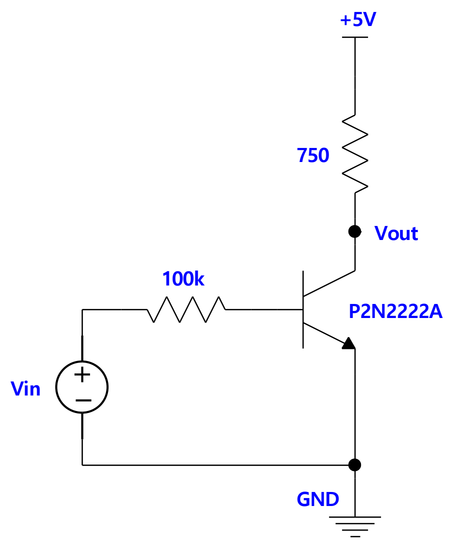
Experiment 1: Transistor Analysis (~40 mins)

Follow the steps below to study the characteristics of an NPN transistor (PN2222A).

Step 1: Switch off the power supply and turn all the knobs fully anticlockwise.

Step 2: Construct the below circuit on the breadboard. Connect the Master Channel of the power supply to +5V of the circuit, and the Slave Channel to V_{in}. Remember that both –ve terminals must connect to the Emitter (E) of the transistor as GND as shown in the figure below (You may refer to Tutorial notes for the breadboard arrangement).





Step 3: Switch on the power supply. Set the current limits for both channels as in Lab1.

Step 4: Set Master Channel to 5V. Set V_{in} (Slave Channel) = 0V. Slowly increase V_{in} from 0V to 5V, 0.2V at a time. Measure V_{out} and fill in the table in the summary sheet.

Q1: Complete the table in the summary sheet.

****** TA Check 1: Show your table record of V_{out} to your TA.**

Q2: Use the data you record in Q1 to make a **line chart** of V_{out} against V_{in} and fill in the summary sheet.

Below instructions are for your reference. Drawing by hand is also acceptable.

https://www.edrawmax.com/line-graph/how-to-make-a-line-graph-in-word/

Notice the decreasing of V_{out} along with an increasing V_{in} from the **line chart**. With such characteristic, transistor can be used to construct inverter. An inverter is a device outputs a voltage representing the opposite logic-level to its input (from low to high, or from high to low). In this lab, you will need to use an inverter IC (74HC14) for the ease of controlling the DC motor rotation directions.

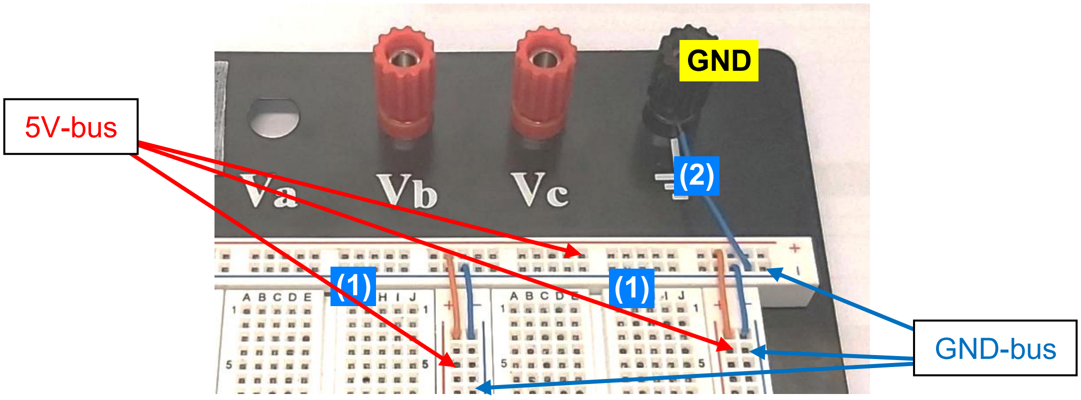
After you've done Experiment 1, remove all of the components from your breadboard.

Experiment 2: H-Bridge Motor Driver (~1 hour)

Starting from now, you are going to construct circuits using ICs for **final project** use.

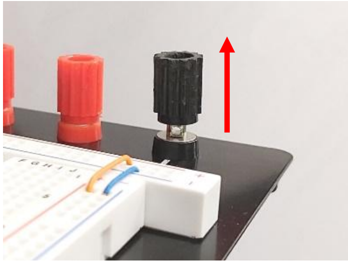
Most of the components we use require the same supply voltage of **5V**. For the tidiness of your breadboard, you should prepare **5V** and **GND** connections using the bus strips (bus strips: refer to your Lab1 manual, on page 2).

Step 1: On your breadboard, connect all the rows of bus trips with the same color. All the rows with red lines will be used as a 5V connection (5V-bus), while all the rows with blue lines will be used as a GND connection (GND-bus).

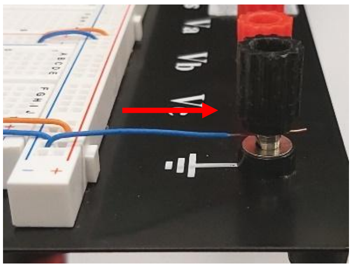


Note: If there is no color marking on the bus strip, you may want to draw red lines and blue lines on the bus strip according to the photo.

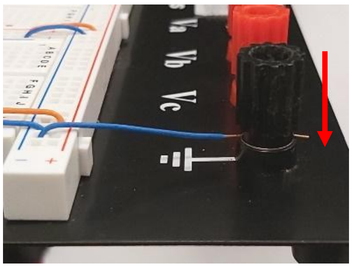
Step 2: Connect a wire from the GND-bus of the bus strip to the **black** binding post.



1. Unscrew the plastic housing



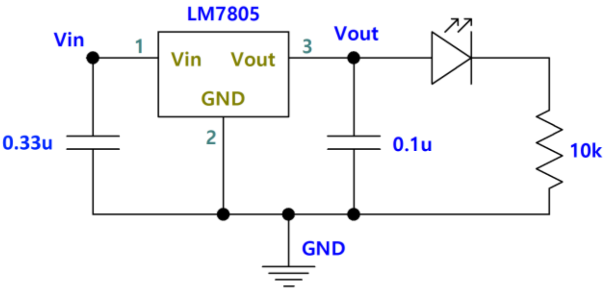
2. Insert wire into the hole. Note the openings of the wire is ~15mm.



3. Screw tight the housing. Make sure that the wire's insulator is not grabbed.

The H-bridge Motor driver circuit requires two different values of power source. Therefore in this experiment, we will need to use the regulator circuit you constructed in Lab3.

Step 3: Re-construct the LM7805 Voltage Regulator circuit on your breadboard.

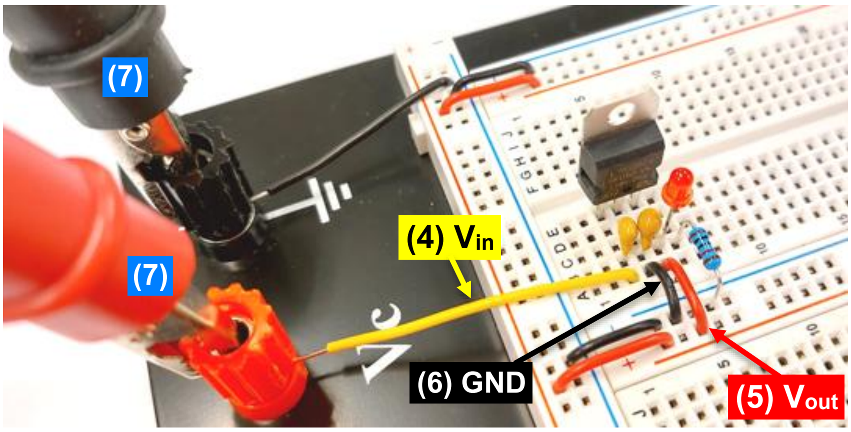


(the markings is facing up)

Step 4: Connect **V_{in}** (will be given to 12V source) to the **red** binding post.

Step 5: Connect **V_{out}** (the generated 5V from LM7805) to the 5V-bus on your breadboard.

Step 6: Connect **GND** to the GND-bus on your breadboard.



The arrangement of components are for reference only. The main goal should be the correct connections and the tidiness for easy debugging.

Step 7: Set the power supply to 0V for now and connect its "+" and "-" terminals to the binding posts to give **V_{in}**.

Step 8: Slowly increase **V_{in}** from 0V to 12V, use a multimeter to verify that **V_{out}** is 5V.

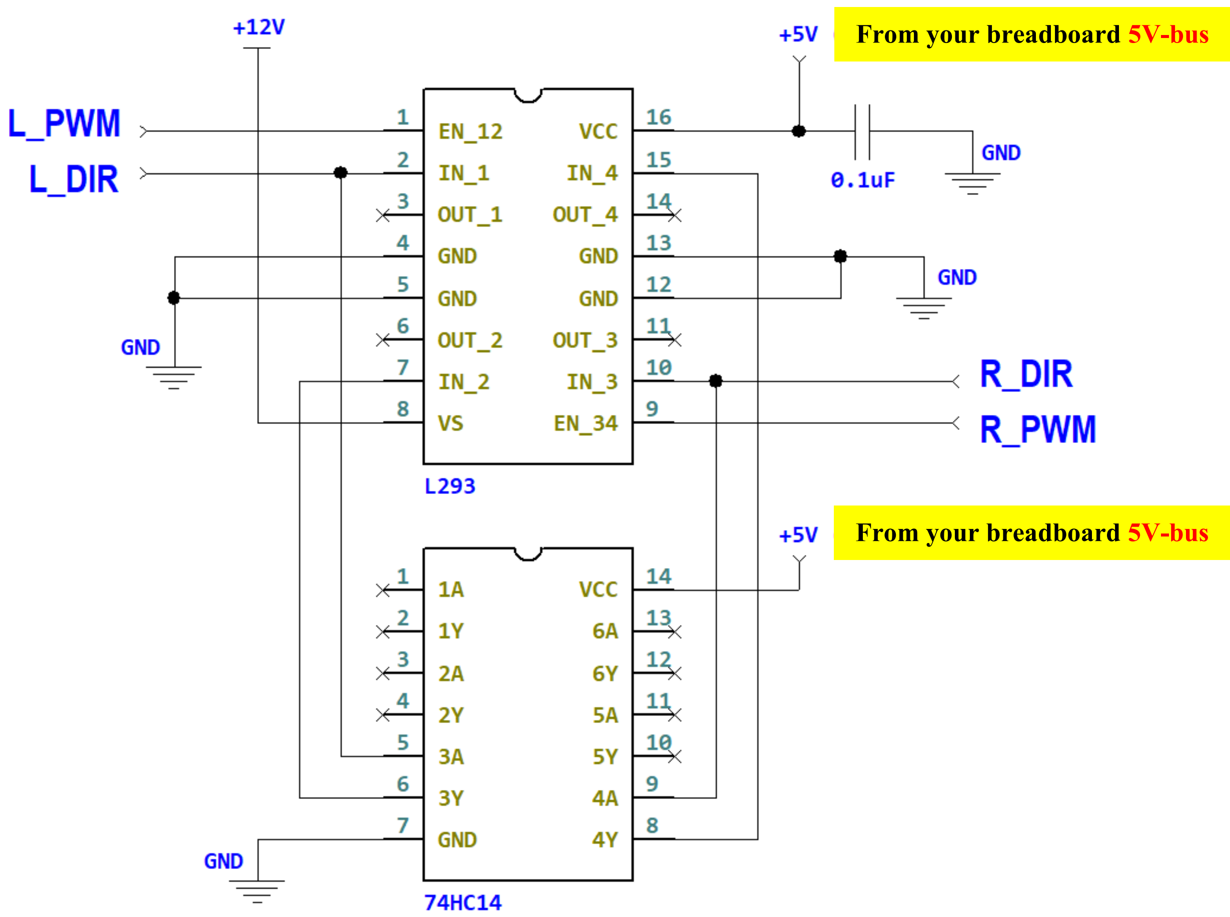
****** TA Check 2.1: Show your TA that you have a regulated 5V from 12V, and have 5V in each 5V-bus on your breadboard.**

Step 9: Connect a driver circuit using L293 and 74HC14 as shown in the figure below.

For this circuit simulation, you need to put your motor driver circuit below the LM7805 regulator circuit on the breadboard (You may refer to Tutorial notes for the breadboard arrangement).

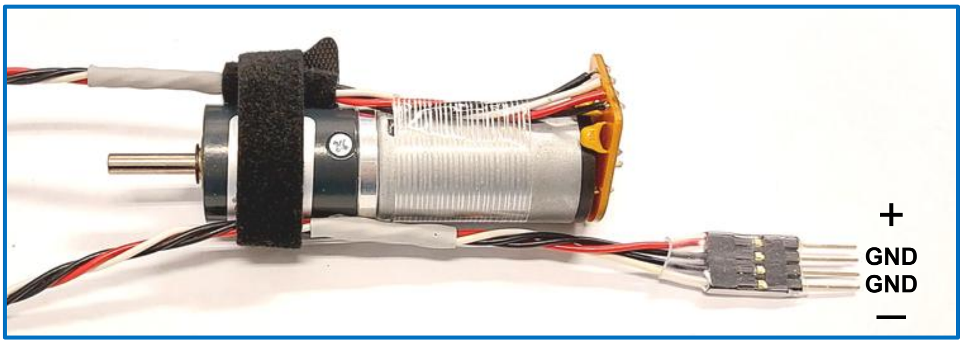
Be clear about where the 12V and 5V (LM7805 input V_{in} and output V_{out}) are on your breadboard.

Do NOT mix up the 12V and 5V, so you won't burn anything.

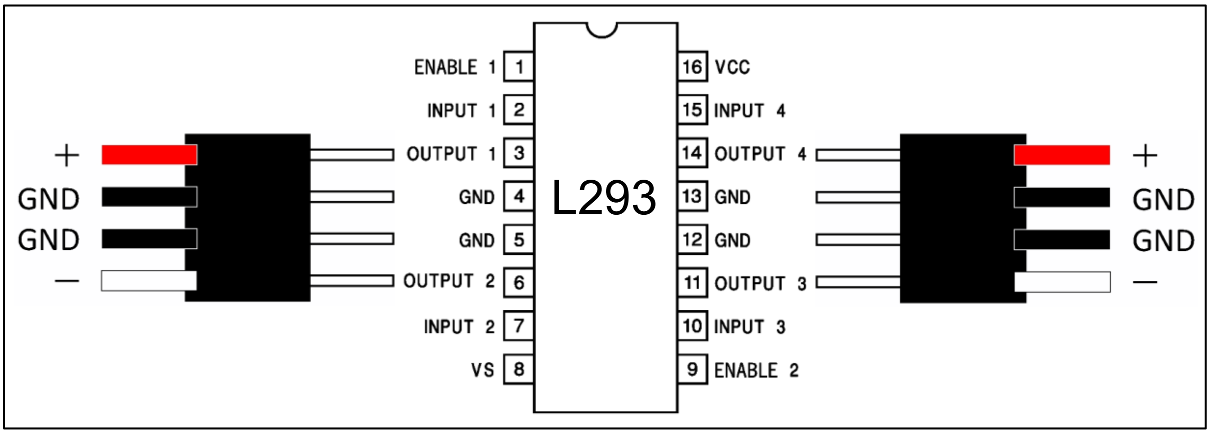


Remember to power **74HC14** by connecting its **pin 14** to your breadboard **5V-bus** and **pin 7** to your breadboard **GND-bus**.

Notice that your given DC motor has the header **as shown below**:



Step 10: Connect two DC Motors to your L293 (H-bridge) circuit. You could easily insert the four terminals of your left motor (through the breadboard row holes) to the left side of L293 (to **pin 3, 4, 5 & 6**), and insert the right motor to the right side of L293 (to **pin 11, 12, 13 & 14**).



Step 11: Connect **Pin 1 (L_PWM)** and **Pin 9 (R_PWM)** of L293 (the PWM speed control) to your breadboard **5V**.

Step 12: Connect **Pin 2 (L_DIR)** and **Pin 10 (R_DIR)** of L293 (the DIR direction control) to your breadboard **5V** **for now**.

****TA Check 2.2: Demo to your TA that,**

1) **when Pin 2 = Pin 10 = 5V (your breadboard 5V-bus), the two motors can turn;**
2) **by changing the connections of Pin 2 & 10 from 5V to 0V (from your breadboard 5V-bus to GND-bus), the two motors can turn in the opposite direction.**

This Experiment demonstrates how to use **Dual H-Bridge Motor Driver IC L293** to control the rotation of a DC motor.

- *Pin 2* changes the polarity (**rotating direction**) between pins 3 and 6.
- *Pin 10* changes the polarity (**rotating direction**) between pins 11 and 14.

This also reveals L293 can control two independent motors at the same time.

KEEP your circuits (LM7805, L293 & 74HC14) on the given breadboard as you will use them at future labs!!!

Remember to clean up your bench and return all the components, breadboard, tools, and cable box! A messy table will cost 1 point!